

Two-Tracer Redshift-Space Power Spectra in FOLPS

Theory, implementation, and validation notes for equal-time cross spectra

FOLPS cross-power development notes

July 25, 2026

Abstract

These notes describe the equal-time two-tracer redshift-space power-spectrum extension implemented in FOLPS. The main motivating application is the cross-correlation of distinct DESI tracers in an overlapping redshift interval, with $\text{LRG3} \times \text{ELG1}$ as the concrete use case. The illustrative numerical figures use LRG-like and ELG-like parameter choices only; they are not DESI best fits. The derivation keeps the two tracer endpoints ordered until the one-loop contractions have been polarized, identifies the exact full biased A -term endpoint factors, and records the implementation and validation checks used by the current branch.

Contents

1	Motivation and scope	2
2	Redshift-space definition	2
3	Bias conventions	3
4	One-loop decomposition	3
5	Density and velocity loops	4
6	The full biased A term	4
6.1	Endpoint exchange	4
6.2	Reduced A term	5
6.3	Additional full- A rows	5
7	D and G terms	5
8	EFT and stochastic terms	6
8.1	Phenomenological cross damping	6
9	Why the geometric mean is not enough	6
10	IR resummation and AP mapping	8
11	Implementation architecture	8
12	Validation	11

13 Numerical illustrations	12
14 Limitations and future work	12
15 Reference metadata	12

1 Motivation and scope

Overlapping galaxy samples contain information that is not exhausted by their two auto spectra. If tracer A and tracer B occupy a common redshift interval, the same large-scale matter and velocity fields source both samples, while their bias expansions, stochastic fields, and pair-level EFT parameters remain different. The cross spectrum P_{AB} probes this shared response directly. It is also the object that enters the covariance matrix of the combined two-tracer data vector.

The cross spectrum is not generally reconstructible from P_{AA} and P_{BB} . The deterministic linear limit has $P_{AB}^L = \sqrt{P_{AA}^L P_{BB}^L}$, but one-loop bias contractions are bilinear in the two endpoint bias vectors, EFT counterterms are pair-level renormalized parameters, and cross stochasticity is not fixed by either auto spectrum. A direct one-loop calculation is therefore required.

The first FOLPS implementation is deliberately narrow. Throughout this note we assume:

- equal time and one common effective redshift for the pair;
- common matter and velocity fields, with no tracer velocity bias;
- the plane-parallel approximation and a common line of sight $\hat{\mathbf{n}}$;
- even power-spectrum multipoles only;
- the standard EFT redshift-space power spectrum;
- optional FolpsD phenomenological cross damping using the prescriptions in Sec. 8.1;
- no window convolution, covariance, likelihood, or redshift averaging.

This implementation branch is based directly on the repository `main` branch. It is not based on the separate `adematti-damping` development branch, and it does not claim feature parity with that branch. Damping support is limited to functionality present and tested on this `main`-based branch. Reconciling this work with other damping-development branches is outside the present scope and is not a scientific dependency of the cross-spectrum theory described here.

Under these assumptions the existing FOLPS FFTLog loop tables are sufficient. The extension reuses the matter, velocity, bias, A , D , G , no-wiggle, and IR-resummation rows, replacing each single-tracer polynomial by the correctly polarized two-tracer polynomial.

2 Redshift-space definition

Let $\hat{\mathbf{n}}$ be the fixed line-of-sight direction and define $\mu = \hat{\mathbf{k}} \cdot \hat{\mathbf{n}}$. The redshift-space cross spectrum is defined by

$$\langle \delta_A^s(\mathbf{k}) \delta_B^s(\mathbf{k}') \rangle = (2\pi)^3 \delta_D(\mathbf{k} + \mathbf{k}') P_{AB}^s(k, \mu). \quad (1)$$

The exact equal-time plane-parallel generating function is

$$(2\pi)^3 \delta_D(\mathbf{k}) + P_{AB}^s(\mathbf{k}) = \int d^3\mathbf{r} e^{-i\mathbf{k} \cdot \mathbf{r}} \left\langle [1 + \delta_A(\mathbf{x}_1)] [1 + \delta_B(\mathbf{x}_2)] e^{-i\mathbf{k}_{\parallel} \Delta v_{\parallel} / (aH)} \right\rangle, \quad (2)$$

where $\mathbf{r} = \mathbf{x}_2 - \mathbf{x}_1$, $k_{\parallel} = \mathbf{k} \cdot \hat{\mathbf{n}}$, and

$$\Delta v_{\parallel} = \hat{\mathbf{n}} \cdot [\mathbf{v}(\mathbf{x}_2) - \mathbf{v}(\mathbf{x}_1)]. \quad (3)$$

The fields δ_A and δ_B denote the real-space density contrasts of the two tracers. Their labels are endpoint labels in (2); they must not be identified until after the contractions have been evaluated.

At equal time, with statistical homogeneity, isotropy, parity invariance, and a common plane-parallel line of sight, the spectrum is real and symmetric under endpoint exchange:

$$P_{AB}^s(k, \mu) = P_{BA}^s(k, \mu). \quad (4)$$

The same assumptions give $P_{AB}^s(k, \mu) = P_{AB}^s(k, -\mu)$, so the present implementation projects only even multipoles,

$$P_{\ell}^{AB}(k) = \frac{2\ell + 1}{2} \int_{-1}^1 d\mu \mathcal{L}_{\ell}(\mu) P_{AB}^s(k, \mu), \quad \ell = 0, 2, 4. \quad (5)$$

Odd multipoles can arise once wide-angle, relativistic, unequal-time, or selection-asymmetry effects are retained; those effects are outside this first FOLPS cross-power path.

3 Bias conventions

The public prior-document input basis is

$$b_1, \quad b_2, \quad b_{K^2}, \quad b_{\text{td}}. \quad (6)$$

The current FOLPS power-spectrum API maps these four entries to the canonical internal basis

$$b_{s^2} = 2b_{K^2}, \quad b_{3\text{nl}} = -\frac{32}{21} \left(b_{K^2} + \frac{2}{5} b_{\text{td}} \right). \quad (7)$$

In canonical FOLPS parameters the density field of tracer X is represented schematically as

$$\delta_X = b_1^X \delta + \frac{b_2^X}{2} [\delta^2] + \frac{b_{s^2}^X}{2} [s^2]_{\text{code}} + b_{3\text{nl}}^X O_{3\text{nl}} + \epsilon_X + \dots \quad (8)$$

The explicit factor 1/2 in the b_2 source is the standard local-quadratic convention. The current FOLPS full- A kernels also insert canonical b_{s^2} with an explicit factor 1/2 in the same second-order source block. Consequently, the full biased A cross term has one factor 1/2 from the second-order source convention and a second factor 1/2 from polarizing an endpoint-summed auto row. This is the origin of the factors 1/4 in Sec. 6.

4 One-loop decomposition

FOLPS organizes the equal-time cross spectrum as

$$\begin{aligned} P_{AB}^s(k, \mu) = & P_{\delta_A \delta_B} + f_0 \mu^2 (P_{\delta_A \theta} + P_{\delta_B \theta}) + f_0^2 \mu^4 P_{\theta \theta} \\ & + A_{AB} + D_{AB} + G_{AB} + P_{\text{ctr}, AB} + P_{\text{stoch}, AB}. \end{aligned} \quad (9)$$

Here θ is the common velocity-divergence field in the FOLPS normalization, f_0 is the large-scale growth rate, and $f(k)$ is the scale-dependent growth rate used by the FOLPS tables. At linear order,

$$P_{K, AB} = (b_1^A + f(k) \mu^2) (b_1^B + f(k) \mu^2) P_L(k). \quad (10)$$

The terms in (9) beyond the cross-Kaiser spectrum are the one-loop density, velocity, redshift-space mode-coupling, EFT, and stochastic pieces.

5 Density and velocity loops

The biased density-density loop polynomial is obtained by bilinearizing the single-tracer FOLPS auto-spectrum polynomial:

$$\begin{aligned}
P_{\delta_A \delta_B}^{\text{loop}} &= b_1^A b_1^B P_{\delta\delta}^{\text{loop}} + (b_1^A b_2^B + b_2^A b_1^B) P_{b_1 b_2} \\
&\quad + (b_1^A b_{s^2}^B + b_{s^2}^A b_1^B) P_{b_1 s^2} + b_2^A b_2^B P_{b_2 b_2} \\
&\quad + (b_2^A b_{s^2}^B + b_{s^2}^A b_2^B) P_{b_2 s^2} + b_{s^2}^A b_{s^2}^B P_{s^2 s^2} \\
&\quad + (b_1^A b_{3\text{nl}}^B + b_{3\text{nl}}^A b_1^B) \sigma_3^2 P_L.
\end{aligned} \tag{11}$$

For either tracer $X \in \{A, B\}$,

$$P_{\delta_X \theta}^{\text{loop}} = b_1^X P_{\delta\theta}^{\text{loop}} + b_2^X P_{b_2 \theta} + b_{s^2}^X P_{s^2 \theta} + b_{3\text{nl}}^X \frac{f(k)}{f_0} \sigma_3^2 P_L. \tag{12}$$

The velocity-velocity loop $P_{\theta\theta}^{\text{loop}}$ is common to both tracers. Equations (11) and (12) use only loop rows that already exist in the FOLPS tables; no new FFTLog matrices are introduced.

6 The full biased A term

The A term is the part of the redshift-space expansion with one pairwise velocity insertion and one ordered bispectrum. Define

$$S_X(\mathbf{q}) = \delta_X(\mathbf{q}) + f_0 \mu_q^2 \theta(\mathbf{q}) \tag{13}$$

and the ordered bispectrum

$$(2\pi)^3 \delta_D(\mathbf{k}_1 + \mathbf{k}_2 + \mathbf{k}_3) B_{\sigma; AB}(\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3) = \langle \theta(\mathbf{k}_1) S_A(\mathbf{k}_2) S_B(\mathbf{k}_3) \rangle. \tag{14}$$

The TNS/FOLPS A term can be written as

$$A_{AB}(k, \mu) = k \mu f_0 \int \frac{d^3 \mathbf{p}}{(2\pi)^3} \frac{p_{\parallel}}{p^2} [B_{\sigma; AB}(\mathbf{p}, \mathbf{k} - \mathbf{p}, -\mathbf{k}) - B_{\sigma; AB}(\mathbf{p}, \mathbf{k}, -\mathbf{k} - \mathbf{p})]. \tag{15}$$

6.1 Endpoint exchange

The Stage 1B endpoint audit resolves the central matrix question. A P22-type FFTLog scalar has the form

$$S[x, M, y] = K^3 \sum_{ij} x_i M_{ij} y_j. \tag{16}$$

Endpoint exchange is the loop-variable transformation $\mathbf{p} \leftrightarrow \mathbf{k} - \mathbf{p}$ together with exchange of the two endpoint spectra. Therefore the physical exchange is

$$x^T M y \longrightarrow y^T M^T x = x^T M y. \tag{17}$$

The exchanged endpoint is not represented by $x^T M^T y$. That object transposes the kernel while leaving the endpoint vectors attached to the wrong endpoints, and is generally different when the vectors differ. Equation (17) is algebraic, not Hermitian; no complex conjugation is involved.

6.2 Reduced A term

With `A_full=False`, the exact cross polynomial is

$$\begin{aligned} A_{AB}^{\text{reduced}} &= b_1^A b_1^B f_0 \mu^2 I_{1udd,1} \\ &\quad + \frac{b_1^A + b_1^B}{2} f_0^2 (\mu^2 I_{2uud,1} + \mu^4 I_{2uud,2}) \\ &\quad + f_0^3 (\mu^4 I_{3uuu,2} + \mu^6 I_{3uuu,3}). \end{aligned} \quad (18)$$

The mixed I_{2uud} rows are endpoint-summed auto rows, so they are polarized by $(b_1^A + b_1^B)/2$. The $I_{1udd,1}$ and pure velocity rows are already symmetric under endpoint exchange.

6.3 Additional full-A rows

With `A_full=True`, the biased second-order rows add

$$\begin{aligned} \Delta A_{AB}^{b_2} &= \frac{b_2^A b_1^B + b_1^A b_2^B}{4} f_0 \mu^2 I_{1udd,1}^{b_2} \\ &\quad + \frac{b_2^A + b_2^B}{4} f_0^2 (\mu^2 I_{2uud,1}^{b_2} + \mu^4 I_{2uud,2}^{b_2}), \end{aligned} \quad (19)$$

and

$$\begin{aligned} \Delta A_{AB}^{s^2} &= \frac{b_{s^2}^A b_1^B + b_1^A b_{s^2}^B}{4} f_0 \mu^2 I_{1udd,1}^{s^2} \\ &\quad + \frac{b_{s^2}^A + b_{s^2}^B}{4} f_0^2 (\mu^2 I_{2uud,1}^{s^2} + \mu^4 I_{2uud,2}^{s^2}). \end{aligned} \quad (20)$$

Each factor 1/4 is the product of two independent factors: 1/2 from the second-order source convention and 1/2 from polarizing an endpoint-summed auto row. In the auto limit $A = B$, (19) and (20) reduce exactly to the canonical FOLPS auto coefficients.

7 D and G terms

The cross D term uses the same row basis as the auto spectrum. Define

$$\begin{aligned} D_2 &= \mu^2 I_{2uudd,1}^D + \mu^4 I_{2uudd,2}^D, \\ D_3 &= \mu^2 I_{3uuud,1}^B + \mu^4 I_{3uuud,2}^D + \mu^6 I_{3uuud,3}^D, \\ D_4 &= \mu^2 I_{4uuuu,1}^B + \mu^4 I_{4uuuu,2}^D + \mu^6 I_{4uuuu,3}^D + \mu^8 I_{4uuuu,4}^D. \end{aligned} \quad (21)$$

Then

$$D_{AB} = b_1^A b_1^B f_0^2 D_2 + \frac{b_1^A + b_1^B}{2} f_0^3 D_3 + f_0^4 D_4. \quad (22)$$

The D_3 coefficient is polarized because the single density insertion can be assigned to either endpoint.

The G term is proportional to the cross-Kaiser polynomial in the current FOLPS convention:

$$G_{AB} = -(k\mu f_0)^2 \sigma_{2w} [b_1^A b_1^B P_L + (b_1^A + b_1^B) f_0 \mu^2 P_{\delta\theta}^L + f_0^2 \mu^4 P_{\theta\theta}^L], \quad (23)$$

where $P_{\delta\theta}^L = (f(k)/f_0)P_L$ and $P_{\theta\theta}^L = (f(k)/f_0)^2 P_L$.

8 EFT and stochastic terms

The first cross implementation uses explicit pair-level nuisance parameters:

$$P_{\text{ctr},AB} = (\alpha_0^{AB} + \alpha_2^{AB}\mu^2 + \alpha_4^{AB}\mu^4)k^2 P_L, \quad (24)$$

$$P_{\text{ctr,NLO},AB} = \tilde{c}^{AB}(k\mu f_0)^4 \sigma_{2w}^2 P_{K,AB}, \quad (25)$$

$$P_{\text{stoch},AB} = P_{\text{shot}}^{AB} [\alpha_{\text{shot},0}^{AB} + \alpha_{\text{shot},2}^{AB}(k\mu)^2]. \quad (26)$$

These parameters are not inferred from the two auto-spectrum nuisance vectors. The public cross API requires them explicitly. The stochastic term is not identified with a Poisson-overlap prediction; P_{shot}^{AB} is a pair-level stochastic normalization in the current parameterization. The implementation does not currently impose a positivity-preserving covariance parametrization for the joint AA , AB , BB model.

8.1 Phenomenological cross damping

For `model="FOLPSD"` the cross-power API supports the same production power-spectrum damping names as the auto path, `"exp"`, `"lor"`, and `"vdg"`. The public keyword `cross_damping_mode` chooses how the cross damping factor is sourced. The default is `"single"`,

$$W_{AB}^{\text{single}}(k, \mu) = W(k, \mu; X_{\text{FoG}}^{AB}), \quad (27)$$

where X_{FoG}^{AB} is the final entry of the explicit pair-level `cross_nuisance` vector. This introduces one damping nuisance parameter for the measured cross spectrum.

The alternative mode is `"geometric"`,

$$W_{AB}^{\text{geometric}}(k, \mu) = \sqrt{W(k, \mu; X_{\text{FoG}}^A)W(k, \mu; X_{\text{FoG}}^B)}. \quad (28)$$

Here X_{FoG}^A and X_{FoG}^B are read from the two tracer parameter arrays after each has been converted through its own bias convention. The pair-level X_{FoG}^{AB} remains present in `cross_nuisance` for API compatibility, but it is ignored in this mode. The geometric prescription preserves exchange symmetry because swapping the tracers only interchanges the two factors under the square root. When $X_{\text{FoG}}^A = X_{\text{FoG}}^B$, it reduces to the corresponding auto damping factor.

The damping placement is intentionally identical to the current auto FolpsD implementation. In the production non-marginalized code, W multiplies the one-loop pair contraction assembled inside `PloopSPTs_cross`. The IR-resummed linear Kaiser term, the standard counterterms, the NLO $\tilde{c}$ counterterm, and the stochastic term are not multiplied by this phenomenological damping factor under the current `Winfty_all=False` convention. Counterterms and stochastic terms retain the pair-level parameterization in Eqs. (24)–(26).

The two damping prescriptions are alternative nuisance models. Neither is claimed to be uniquely selected by first principles; mocks and data should be used to decide whether the extra tracer-level structure in the geometric mode is useful.

9 Why the geometric mean is not enough

In the deterministic linear limit,

$$P_{AB}^L = \sqrt{P_{AA}^L P_{BB}^L}, \quad (29)$$

Table 1: Current FOLPS matrix and loop-table mapping for the A rows. The suffix “_b” on vectors denotes the auxiliary FFTLog bias exponent used by FOLPS, not tracer B .

Function	Matrix or vector pieces	Left vector	Right vector	Loop-table row
$I_{1udd,1}$	M_{11}^{tAfp} plus $M_{11}^{afk}, M_{11}^{afp}$	vecf	vec	I1udd_1
$I_{2uud,1}$	$M_{12}^{tAfkmpfp}$ plus $M_{12}^{afkfp}, M_{12}^{afpfp}$	vecf	vecf	I2uud_1
$I_{2uud,2}$	$M_{22}^{tAfkmpfp}, M_{22}^{tAfpfp}$ and P13 pieces	vecf, vecff	vecf, vec	I2uud_2
$I_{3uuu,2}$	$M_{23}^{tAfkmpfpfp}$ plus P13 inherited pieces	vecff	vecf	I3uuu_2
$I_{3uuu,3}$	$M_{33}^{tAfkmpfpfp}$ plus $M_{33}^{afkfkfp}, M_{33}^{afkfpfp}$	vecff	vecf	I3uuu_3
$I_{1udd,1}^{b2}$	$M_{11,b2}^{tAfp}$	vecf_b	vec_b	I1udd_1_b2
$I_{2uud,1}^{b2}$	$M_{12,b2}^{tAfkmpfp}$	vecf_b	vecf_b	I2uud_1_b2
$I_{2uud,2}^{b2}$	$M_{22,b2}^{tAfkmpfp}$	vecf	vecf	I2uud_2_b2
$I_{1udd,1}^{s2}$	$M_{11,bs2}^{tAfp}$	vecf_b	vec_b	I1udd_1_bs2
$I_{2uud,1}^{s2}$	$M_{12,bs2}^{tAfkmpfp}$	vecf_b	vecf_b	I2uud_1_bs2
$I_{2uud,2}^{s2}$	$M_{22,bs2}^{tAfkmpfp}$	vecf	vecf	I2uud_2_bs2

Illustrative FolpsD cross damping modes, VDG

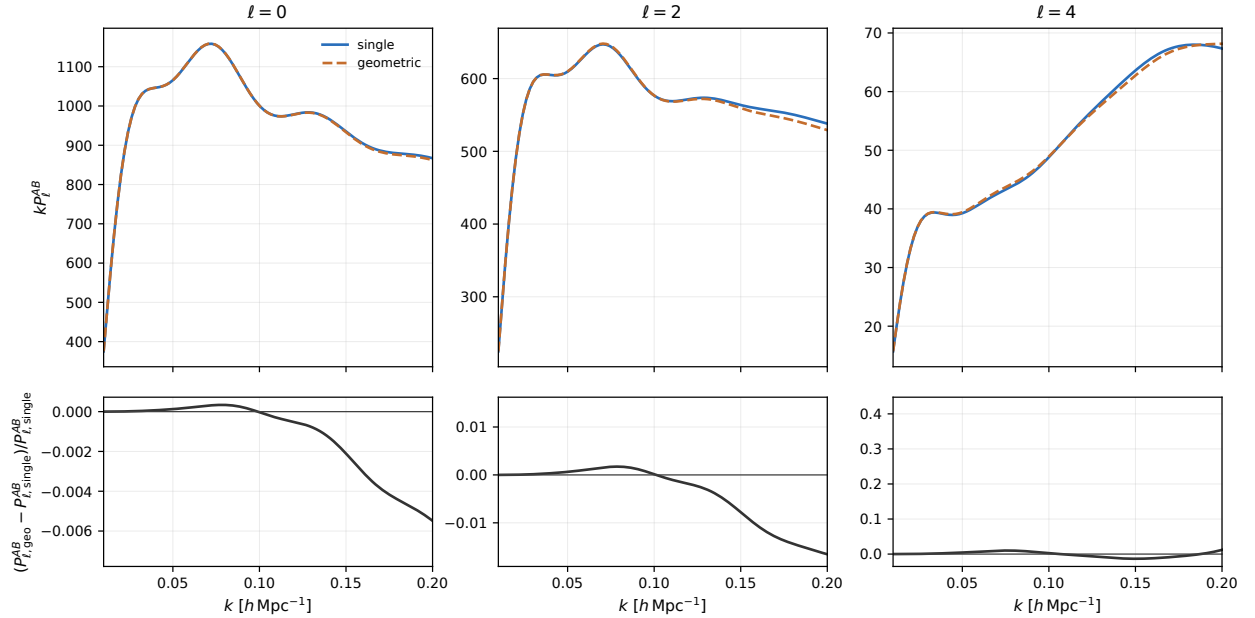


Figure 1: Illustrative FolpsD cross damping comparison for VDG damping. The single mode uses one pair-level X_{FoG}^{AB} , while the geometric mode uses the two tracer-level damping factors. The lower panels show the fractional difference relative to the single prescription.

because all spectra are products of the same linear matter power spectrum and linear endpoint responses. Beyond that limit, write schematically

$$P_{XX} = P_{XX}^L + P_{XX}^{1\text{-loop}}. \quad (30)$$

Then

$$\sqrt{P_{AA}P_{BB}} \simeq \sqrt{P_{AA}^L P_{BB}^L} \left[1 + \frac{1}{2} \frac{P_{AA}^{1\text{-loop}}}{P_{AA}^L} + \frac{1}{2} \frac{P_{BB}^{1\text{-loop}}}{P_{BB}^L} \right], \quad (31)$$

which averages fractional auto corrections. The true cross loop instead contains bilinear monomials such as $b_2^A b_1^B + b_1^A b_2^B$ and $b_2^A b_{s^2}^B + b_{s^2}^A b_2^B$. Cross EFT and stochastic terms are independent pair-level renormalized parameters. Therefore the geometric mean is at most a diagnostic approximation.

10 IR resummation and AP mapping

The cross implementation keeps the existing FOLPS IR-resummation structure. The displacement damping is shared because the two tracers live in the same matter and velocity fields. Only the tracer-dependent polynomials are replaced by their cross forms. In code notation,

$$\begin{aligned} \Sigma_t^2(k, \mu) &= [1 + f_0 \mu^2 (2 + f_0)] \Sigma^2 + (f_0 \mu)^2 (\mu^2 - 1) \Delta \Sigma^2, \\ E(k, \mu) &= \exp[-k^2 \Sigma_t^2(k, \mu)], \end{aligned} \quad (32)$$

and the returned redshift-space spectrum is assembled as

$$\begin{aligned} P_{AB}^{s,\text{IR}} &= P_{K,AB} [P_{\text{nw}} + E(P_L - P_{\text{nw}})(1 + k^2 \Sigma_t^2)] \\ &\quad + E P_{AB}^{1\text{loop}}[\text{wiggle}] + (1 - E) P_{AB}^{1\text{loop}}[\text{nowiggle}]. \end{aligned} \quad (33)$$

The Alcock-Paczynski mapping is also unchanged. With $F = q_{\parallel}/q_{\perp}$,

$$\begin{aligned} k_{\text{true}} &= \frac{k_{\text{obs}}}{q_{\perp}} [1 + \mu_{\text{obs}}^2 (F^{-2} - 1)]^{1/2}, \\ \mu_{\text{true}} &= \frac{\mu_{\text{obs}}}{F} [1 + \mu_{\text{obs}}^2 (F^{-2} - 1)]^{-1/2}, \end{aligned} \quad (34)$$

and the multipoles are multiplied by the Jacobian

$$(q_{\parallel} q_{\perp}^2)^{-1}. \quad (35)$$

11 Implementation architecture

The implementation remains in `folps/folps.py`; no `folpsX.py` module is introduced. The non-marginalized auto and cross spectra both pass through the same private pair-level helper. Auto mode is obtained by setting $A = B$ and using the auto nuisance tuple. Cross mode is activated by passing `pars_b` and an explicit `cross_nuisance` tuple.

The implementation uses small `NamedTuple` containers for the canonical bias and nuisance subsets. The public multipole method accepts `bias_scheme` for tracer A and `bias_scheme_b` for tracer B ; the prior-document convention is converted by (7) before entering the pair contraction. Direct `get_rsd_pkmu` and `get_eft_pkmu` calls use canonical FOLPS parameter ordering.

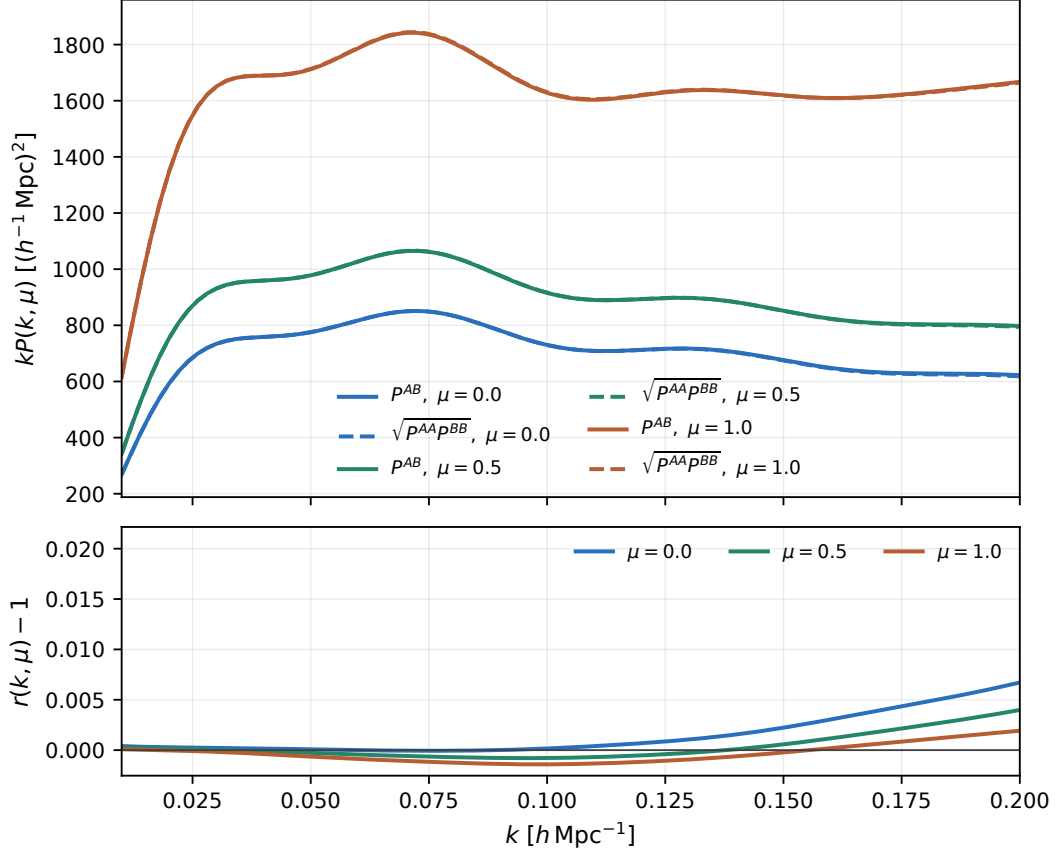


Figure 2: Comparison of the implemented $P^{AB}(k, \mu)$ with the diagnostic geometric mean $\sqrt{P^{AA}(k, \mu)P^{BB}(k, \mu)}$ for three angular slices. The lower panel displays $P^{AB}/\sqrt{P^{AA}P^{BB}} - 1$ only where the geometric mean is real and numerically stable.

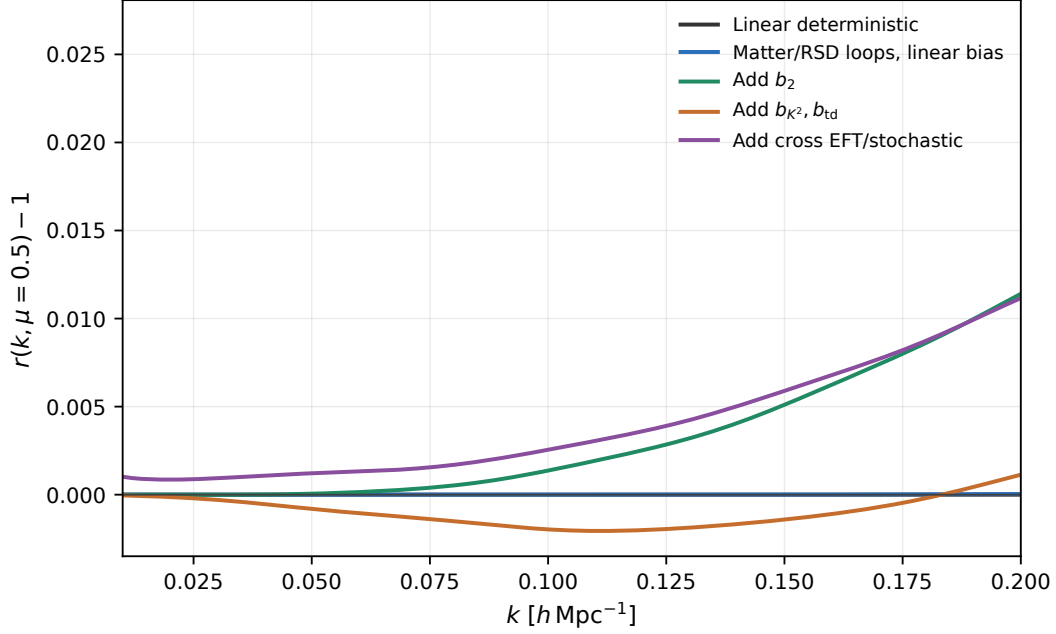


Figure 3: Controlled sector switch for the geometric-mean diagnostic at $\mu = 0.5$. The deterministic linear synthetic table gives $r - 1$ consistent with zero, while matter/RSD loops, nonlinear bias terms, and explicit pair-level nuisance parameters generate departures from the geometric mean.

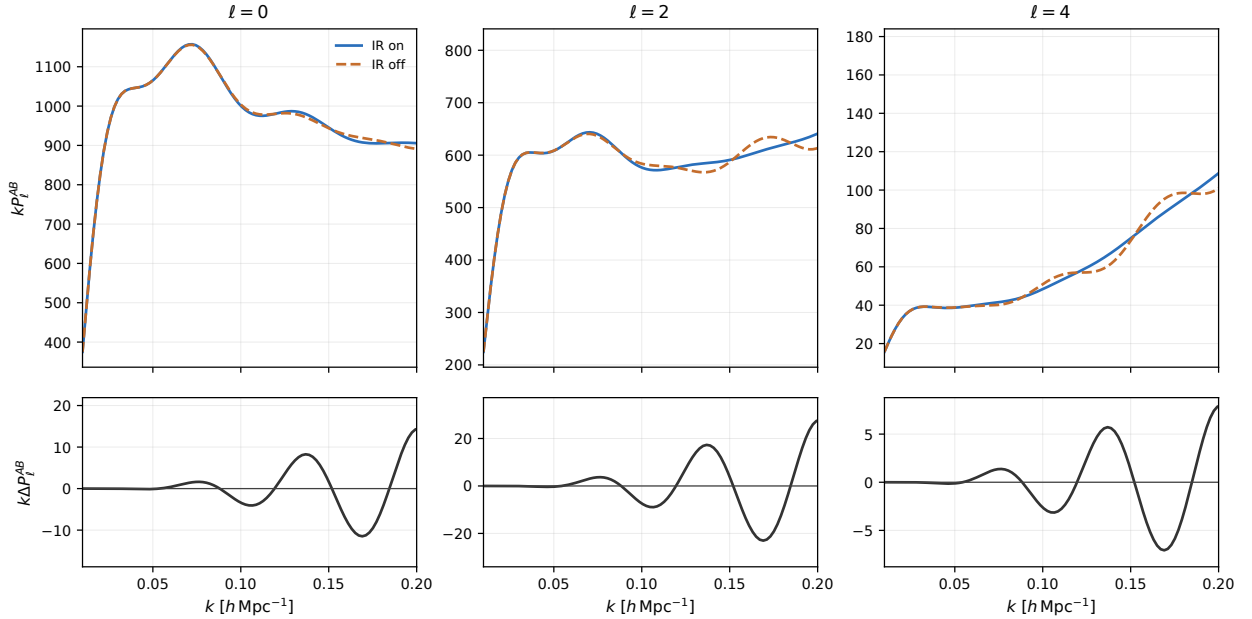


Figure 4: Numerical effect of the existing FOLPS IR resummation on the cross multipoles. The upper row compares the resummed and non-resummed calculations; the lower row shows their absolute difference.

Cross spectra are supported for `model="EFT"` and `model="FOLPSD"`. With `model="FOLPSD"`, the public `cross_damping_mode` keyword accepts `"single"` and `"geometric"`, with `"single"` as the default. The table and `table_now` schemas are reused directly, including the six `A_full=True` rows, the no-wiggle rows, and the IR scalar tail. No matrix file is added or modified. The NumPy and JAX backends share the same source formulas through the module-selected backend namespace; JAX device selection is environment dependent and is not restricted to CPU by the cross-power code.

The FolpsD cross damping path supports only the main-branch damping names `"exp"`, `"lor"`, and `"vdg"`. No additional damping names, fallback aliases, or flags from `adematti-damping` are part of this branch.

```
pars_a = [...]
pars_b = [...]
cross_nuisance_ab = [
    alpha0_ab, alpha2_ab, alpha4_ab, ctilde_ab,
    alphashot0_ab, alphashot2_ab, PshotP_ab, X_FoG_ab,
]

multipoles = RSDMultipolesPowerSpectrumCalculator(model="FOLPSD")
p0_ab, p2_ab, p4_ab = multipoles.get_rsd_pkell(
    kobs, 1.0, 1.0, pars_a, table, table_now,
    bias_scheme="priordoc",
    bias_scheme_b="priordoc",
    pars_b=pars_b,
    cross_nuisance=cross_nuisance_ab,
    damping="vdg",
    cross_damping_mode="single",
    IR_resummation=True,
    ells=(0, 2, 4),
)
```

12 Validation

The independent tests in `folps/test_cross_power_spectrum.py` exercise both algebraic identities and backend behavior:

- linear Kaiser recovery with synthetic tables;
- one-row synthetic loop checks for every density, density-velocity, velocity-velocity, reduced- A , full- A , D , and G row;
- all six full- A rows with asymmetric tracer biases;
- exchange symmetry $P_{AB} = P_{BA}$;
- auto-limit recovery for `A_full=False` and `A_full=True`;
- prior-document/canonical equivalence;
- explicit missing-`cross_nuisance` rejection;

- IR on/off checks;
- FolpsD cross damping checks for "exp", "lor", and "vdg" in both "single" and "geometric" modes;
- AP-remapped multipole checks against direct quadrature;
- NumPy/JAX agreement;
- unchanged auto-spectrum regression against existing outputs.

For the plotted run, the exchange, auto-limit, and prior-document/canonical residuals are exactly zero. In Fig. 6, the 10^{-16} level is only a visualization floor for logarithmic plotting of those zero residuals.

13 Numerical illustrations

The generated figure set uses the public input linear spectrum and the existing `A_full=True`, `use_TNS_model=False` matrix cache. The common settings are `model="EFT"`, `damping=None`, IR resummation on, even multipoles (0, 2, 4), and prior-document bias inputs. All parameter changes in this section are illustrative visibility choices, not priors. The cross damping comparison in Fig. 1 explicitly switches to `model="FOLPSD"` and `damping="vdg"`.

14 Limitations and future work

The current implementation is the first equal-time two-tracer power-spectrum path. The main limitations and natural extensions are:

- effective-redshift modeling and redshift averaging across the true overlap window;
- realistic LRG3 $\times$ ELG1 parameter choices and data-vector definitions;
- cross stochasticity models that enforce positivity of the joint 2×2 tracer covariance;
- field-level counterterm parametrizations that relate auto and cross EFT coefficients;
- survey window convolution and integral-constraint effects;
- covariance and likelihood integration, including desilike interfaces;
- tracer velocity bias;
- odd multipoles, wide-angle terms, and relativistic corrections;
- mock- and data-driven comparisons of the available FolpsD cross damping nuisance prescriptions;
- bispectrum cross-correlations and joint power-spectrum/bispectrum two-tracer analyses.

15 Reference metadata

Reference metadata were checked against the local notes and arXiv records for the four required entries. The original TNS paper is Ref. [1]; the FOLPS and FFTLog massive-neutrino references are Refs. [2, 3]; and the FolpsD reference is Ref. [4].

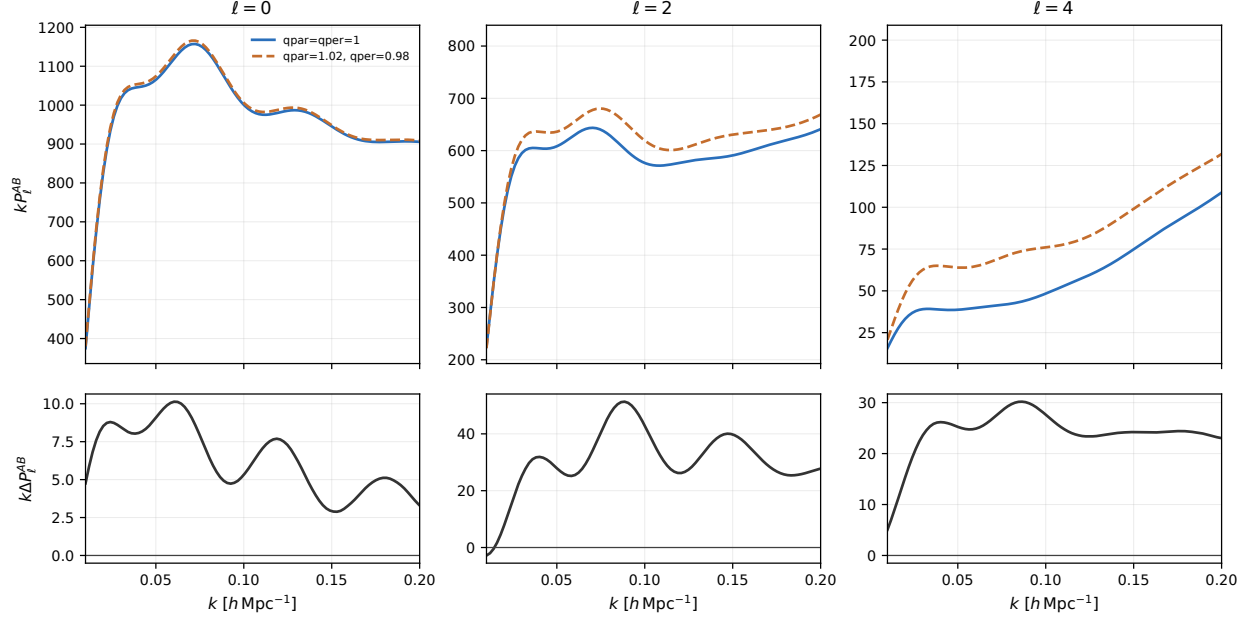


Figure 5: Illustrative Alcock-Paczynski remapping for the cross multipoles. The distorted case uses $q_{\parallel} = 1.02$ and $q_{\perp} = 0.98$ as a numerical demonstration, not as a DESI fiducial choice.

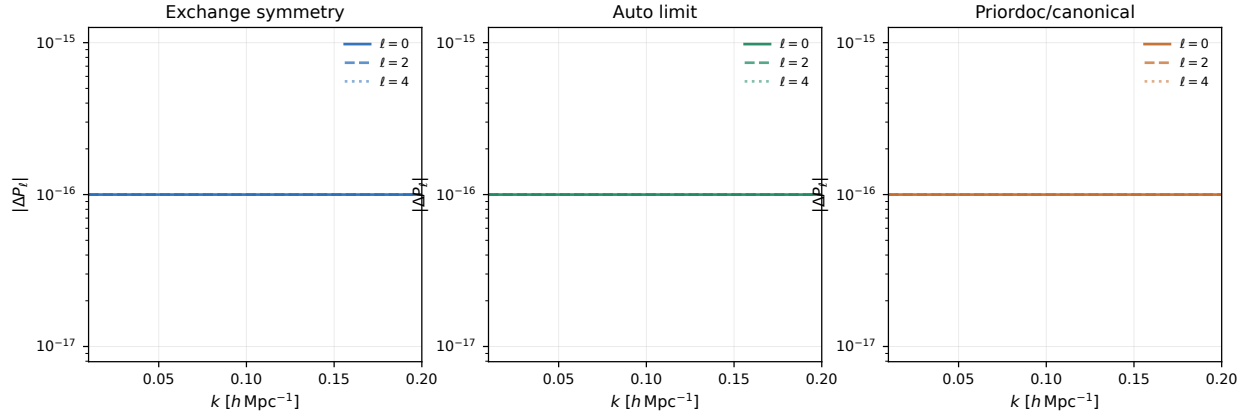


Figure 6: Implementation residuals for exchange symmetry, auto-limit recovery, and prior-document versus canonical input conversion. The logarithmic floor is a display device only and has no physical significance.

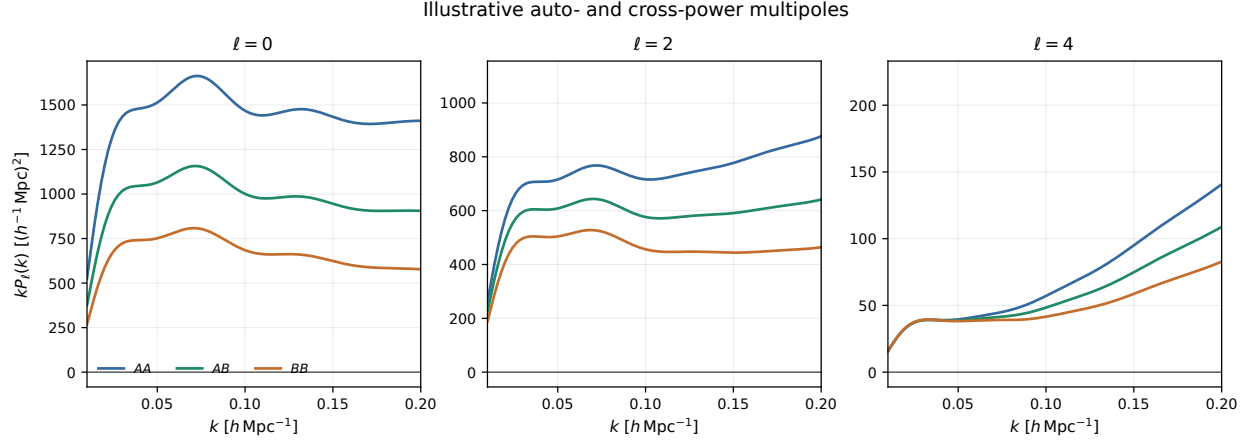


Figure 7: Illustrative auto and cross multipoles for the LRG-like tracer A , the ELG-like tracer B , and their equal-time cross spectrum. The cross multipole is a projected EFT model output and is not constrained to lie between the two auto multipoles at every k and ℓ .

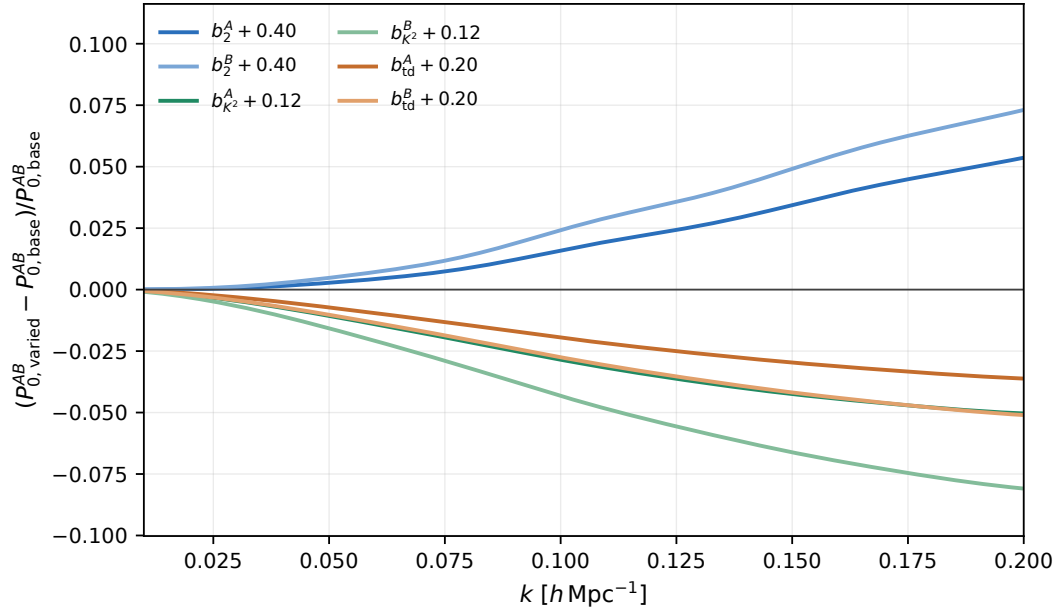


Figure 8: Fractional response of the cross monopole when one nonlinear prior-document bias parameter is varied for tracer A or tracer B . The independent responses illustrate that the cross loop polynomial is genuinely bilinear in the two tracer bias vectors.

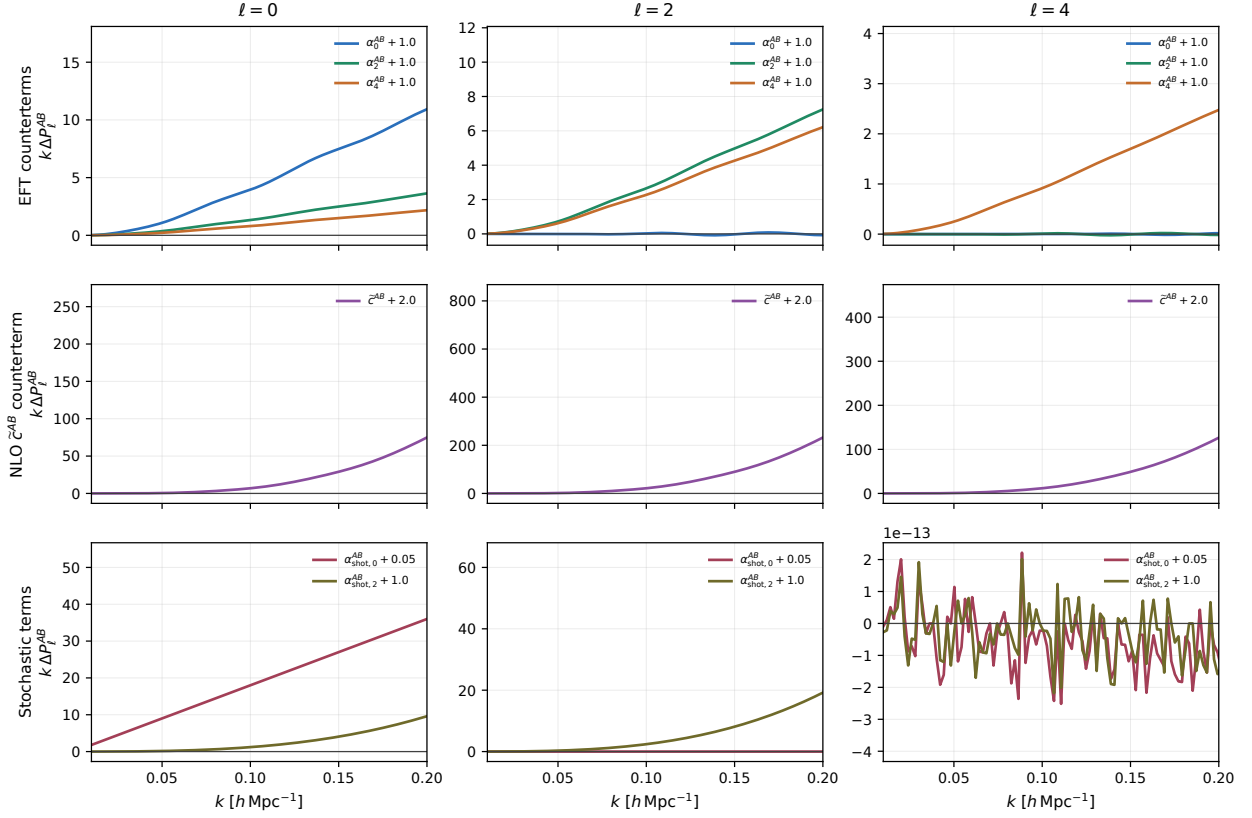


Figure 9: Response of the cross multipoles to explicit pair-level EFT and stochastic parameters. The rows isolate the standard EFT counterterms, the NLO $\tilde{c}^{AB}$ counterterm, and the stochastic terms with independent y-axis ranges so the smaller responses remain visible.

References

- [1] Atsushi Taruya, Takahiro Nishimichi, and Shun Saito. Baryon acoustic oscillations in 2d: Modeling redshift-space power spectrum from perturbation theory. *Physical Review D*, 82:063522, 2010. doi: 10.1103/PhysRevD.82.063522. [arXiv:1006.0699](#).
- [2] Alejandro Aviles, Arka Banerjee, Gustavo Niz, and Zachary Slepian. Clustering in massive neutrino cosmologies via eulerian perturbation theory. *Journal of Cosmology and Astroparticle Physics*, 2021(11):028, 2021. doi: 10.1088/1475-7516/2021/11/028. [arXiv:2106.13771](#).
- [3] Hernan E. Noriega, Alejandro Aviles, Sebastien Fromenteau, and Mariana Vargas-Magana. Fast computation of non-linear power spectrum in cosmologies with massive neutrinos. *Journal of Cosmology and Astroparticle Physics*, 2022(11):038, 2022. doi: 10.1088/1475-7516/2022/11/038. [arXiv:2208.02791](#).
- [4] P. Bansal et al. FolpsD: combining EFT and phenomenological approaches for joint power spectrum and bispectrum analyses, 2026. [arXiv:2604.08895](#).